TABLE OF CONTENTS

1 - INTRODUCTION

1-1 Overview

Note:
Do **not** use this procedure on a 10.X Excite or Linux based system.

This procedure should be used for all models of LCDs that do **not** have a specific procedure written for them, and do **not** appear in the Display Monitor drop-down list in the Install GUI Display Type field.

This procedure covers software versions:

- 9.0, VH1, VH2, VH3, CNV3, CNV4, HFO1, ASP1, ASP2
- AND ALL SOFTWARE VERSIONS ON INDIGO2

If the Monitor Type being installed appears in the list below, use that procedure, which is designed for the monitor. These procedures change frequently. Check the MR Service Engineering Website for the most up-to-date procedures before proceeding.

TABLE 1-1
CURRENT HOST MONITORS

All CRTs -	Setup & Calibration Procedure
NEC 2010 -	Setup & Calibration Procedure
NEC2010X -	Setup & Calibration Procedure
NEC 1850X -	Setup & Calibration Procedure
NEC 1880SX -	Setup & Calibration Procedure
NEC 1990SX -	Setup & Calibration Procedure
EIZO L660 -	Setup & Calibration Procedure
Generic or Basic -	CRT or LCD Monitor not listed? Use this procedure.

1-2 Camera Vendor Participation

This procedure should be performed with the Camera Vendor Field Engineer present, should any camera adjustment be necessary. To optimize customer satisfaction, one of the customer's Filming Specialists should be available for the fine tuning and quality review of the film/LCD display monitor conformance.

2 - MONITOR INSTALLATION

2-1 Installation of LCD Monitor in a Fixed Site Environment

1. Shutdown Signa and perform a safety Lock-Out/Tag-Out (LOTO) of the Operator Workspace. (See Safety procedure to accomplish this.)
2. If necessary, remove the old LCD monitor from the Desktop.
3. Remove the new LCD monitor from its packaging, and set the monitor on the workspace desktop.
4. Rotate the monitor and connect the video and power cables.
5. Remove LOTO from the Operator Workspace, and restore power to the system.
6. Verify that main LCD monitor power is ON. If not, locate and press the power button on LCD. (This button should remain on at all times.)
7. Make sure the monitor is positioned no closer than 16 inches and no further away then 28 inches from the eyes. The optimal distance is 24 inches for either of the monitors.

Note

Allow the monitor to warm up for 20 minutes before performing any adjustments.

8. Boot up Signa.
9. Use the password: **adw2.0** and press **<Enter>**.

2-2 Installation or Replacement of an LCD Monitor in a Mobile Environment

This section pertains to mobile systems only. Replacement of an LCD is performed by removing the screws under the tabletop that hold the existing LCD panel base. Determine the best way to secure the new monitor to the tabletop.

3 - VERIFY AMBIENT LIGHTING CONDITIONS

LCDs are extremely susceptible to viewing angle. All adjustments and image comparisons should be done from exactly the same viewing angle in all directions. Sit directly in front of the monitor at all times.

In the Operator Workspace area:
- Verify there is only sufficient light for safely operating the system.
- Verify that light boxes are not emitting light or are properly masked when not displaying film. This will be a source of excessive glare.

In both Review and Operator Workspace areas:
- Verify that there is no source of glare for reviewing films or setting up the images for film. For example, windows should not allow direct light. (Blinds should be closed).
- Verify that the ambient lighting conditions are adjusted to a minimum level.
- Make sure that the artificial lighting is of the same type.

4 - SETTING LCD MONITOR FREQUENCY RATE AND RESOLUTION

Failure to follow this section will result in Image Ghosting and unnecessary Troubleshooting.

4-1 Setting Line Rate and Resolution

Monitor "Line Rate" or "Refresh Rate" is referred to as Frame Rate and Swap Rate on SGI systems. For simplicity, the term Line Rate will be used in this document.

WARNING!

THIS PROCEDURE NEEDS TO BE DONE EVERY TIME SOFTWARE IS RELOADED. LINE RATE IS NOT CAPTURED ON THE SAVEINFO MOD, AND IS NOT CAPTURED ON THE HEALTH PAGE STATUS OF THE SYSTEM. FAILURE TO SET THE LINE RATE TO 60 HZ FOR ALL LCD HOST MONITORS WILL RESULT IN IMAGE GHOSTING AND CUSTOMER COMPLAINTS.

Note

UNIX/IRIX is case sensitive. Use the commands exactly as shown for root owner.

1. From the Desktop, click on the **C-shell** tool.
2. Switch to root owner and type: **su root** and press **<Enter>**; Type the password.
3. Type: **chkconfig fixed72hz off** and press **<Enter>**.

Note

If the following error displays, a command might be typed incorrectly. Retype the command. If it displays a second time, ignore it and proceed with Step 4.

```
chkconfig: cannot set flag "fixed72hz": No such file or directory
use "chkconfig -f fixed72hz off" to create and set the flag
```

4. Type: **/usr/gfx/setmon –x –w 1280x1024_60**.
5. Type: **exit** and press **<Enter>** to exit root.
6. Type: **exit** and press **<Enter>** to exit the C-shell.
7. Select **System Shutdown** and **OK** to Shutdown.
8. Select **Restart**.
9. Click the *Signa* icon and type the password.
10. Proceed to Section 5, Default Signa Host Gamma Setup (Generic/Basic LCD Monitor).

Note

To check line rate values, refer to Troubleshooting section at the end of this procedure.

5 - DEFAULT SIGNA HOST GAMMA SETUP (GENERIC/BASIC LCD MONITOR)

The Gamma value is modified to optimize the contrast level of the image mid tones to more closely represent the same contrast that is filmed.

Note

At time of System installation:

As of July 28, 2003, all **new** LCD monitors will include a Gamma Software CD-ROM (GE P/N 2376768) that should be used for software versions 9.0 and below. Use this CD-ROM and the supplied instructions to set up gamma for those software versions.

Upon replacement of a failed monitor:

Make sure to also order GE P/N 2376768. This CD-ROM is labeled "MR Signa® Host Monitor Gamma Correction Software" and contains the most up-to-date files required to set up gamma. The software on the CD-ROM is **required** to install a replacement monitor. Follow the installation instructions included with the CD-ROM.

WARNING!

MR SIGNA® HOST MONITOR GAMMA CORRECTION SOFTWARE (GE P/N 2376768) SUPERCEDES ALL OTHER GAMMA FILES THAT MAY BE ON THE VERSION 14 AND 15 LX/8.X SERVICE METHODS CD-ROM.

5-1 Gamma Configuration for All SGI Computers

Using the table below, select the section to use based on your Computer type and Software version:

GAMMA PROCEDURES BY SECTION

TABLE 5-1

Computer Type:	Software Version:	Gamma Setting:
All SGI computers	ALL Software versions 9.0 and below. Setting Gamma <u>with</u> the CD-ROM that came with the monitor.	See **Section 5-1-1.**
Octane 1 and Octane 2	9.1, HFO2, HF03 Setting Gamma on monitors <u>without</u> a CD-ROM	See **Section 5-1-2**.

5-1-1 Setting Gamma with Patch CD-ROM (All Software Versions 9.0 and Below)

Proceed as follows to set the gamma with the patch CD-ROM:

- If installing the monitor for the **first time**, a CD-ROM labeled "MR Signa® Host Monitor Gamma Correction Software" is included in the monitor box.

- If replacing a **failed** LCD monitor, a CD-ROM labeled "MR Signa® Host Monitor Gamma Correction Software" (GE P/N 2376768) should have been ordered along with the replacement monitor.

- Follow the installation instructions included with either CD-ROM to load the gamma software onto the system.

- This CD-ROM should be retained and re-used any time a reload of software occurs on the system.

- If the software version is not known, type: **getver** in a C-shell.

The following option is used to configure a CRT or LCD model that is **not** present in the drop-down list. To set Gamma:

- If replacing the **same** model and make of CRT or LCD and software was **not reloaded**: Gamma changes are unnecessary. Proceed to Section 6, Setting Default Contrast and Brightness for LCD Monitor.
- If replacing the **same** model and make of CRT or LCD and software **was reloaded**: Proceed to Step 1 below.
- If **reloading or loading** these versions of software for the first time and the CRT or LCD on the system has **no** specific procedure or CD-ROM: Finish the software load process and return to Step 1 below.

1. Insert the “MR Signa® Host Monitor Gamma Correction” CD-ROM into the Host CD-ROM/DVD-ROM drive.

2. Follow the instructions included with the CD-ROM to set the gamma for your monitor type.

3. Proceed to Section 6, Setting Default Contrast and Brightness for LCD Monitor.

5-1-2 Setting Gamma (9.1, HF02 and HFO3)

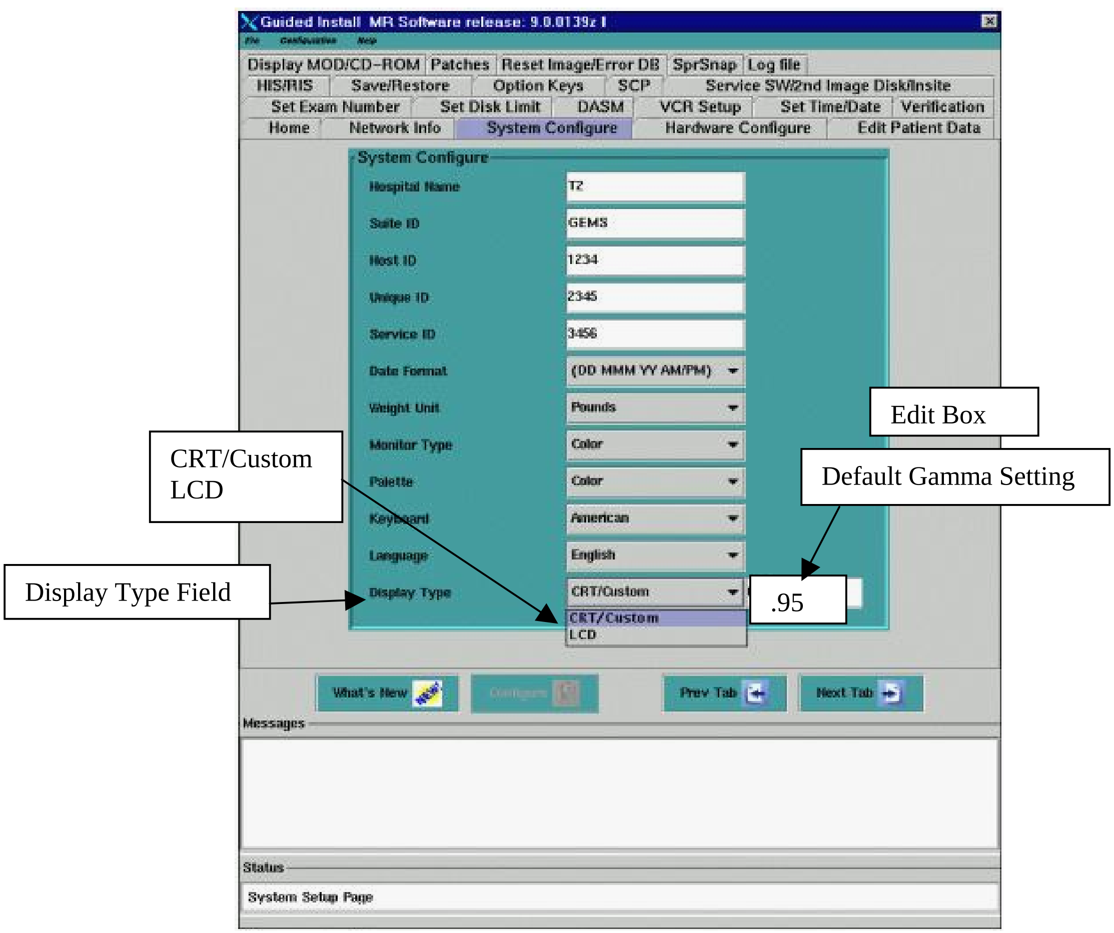
This option is used to configure a CRT or LCD model that is **not** present in the drop-down list.

To set Gamma:

- If replacing the **same** model and make of CRT or LCD and software was **not reloaded**: **No** gamma changes are necessary. Proceed to Section 6, Setting Default Contrast and Brightness for LCD Monitor.
- If replacing the **same** model and make of CRT or LCD and software **was reloaded**: Proceed to Step 1 below.
- If **reloading or loading** these versions of software for the first time and the CRT or LCD on the system has **no** specific procedure or CD-ROM: Finish the software load process and return to Step 1 below.

1. Open Guided Install and select the *System Configure* tab. (See Illustration 5-1.)

2. Locate the *Display Type* field shown in Illustration 5-1.



GUIDED INSTALL SYSTEM CONFIGURE TAB- DISPLAY TYPE FIELD
ILLUSTRATION 5-1

Note

The *Display Type* field has two choices:

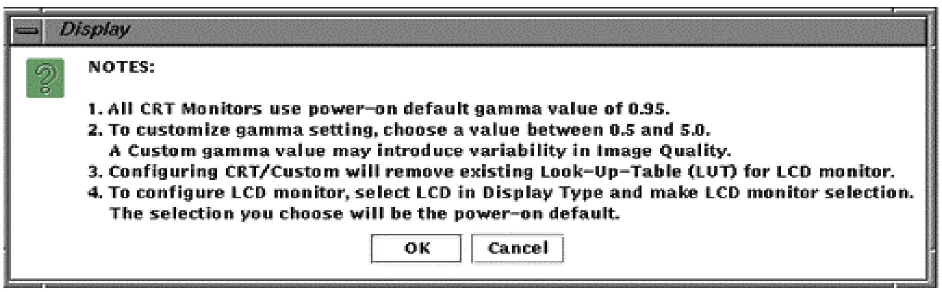
- **CRT/Custom** – Use to set a Custom LCD or Custom CRT Gamma setting.
- **LCD** – Use to configure Gamma for a specific LCD listed in the drop-down box (preferred method for LCDs).

It may be necessary to toggle between the **LCD** and **CRT/Custom** options in the *Display Type* field.

1. If **LCD** displays in the *Display Type* field:

   a. Click on the *Display Type* field and select **CRT/Custom**.

   b. When a pop-up shown in Illustration 5-2 displays, select **[OK]**. (The *Display Type* field should change to CRT/Custom with the cursor in the *Edit* box shown in Illustration 5-1.)



MESSAGE BOX
ILLUSTRATION 5-2

   c. Use the default value of **.95**. If the customer does not want this value, type the desired gamma value into the *Edit* box to the right of the *Display Type* field (shown in Illustraton 5-1) and press **<Enter>**.

Note

A higher number will increase and a lower number will decrease Brightness & Contrast. The Gamma value of any LCD is dependent on the LCD and Display hardware as well as the customer's choice. This can vary from site to site.

   d. Click the **[Configure]** button on the Install GUI to save the changes. (If the gamma value changed, a slight change in the screen intensity will be noticed when Configure is completed.)

   e. When a pop-up displays, select **[OK]**.

   f. Select **File** and **Exit** and **[OK]** on the *Install/Reconfigure* window.

   g. **Reboot** the system. (A reboot is required to activate the changes.)

   h. Proceed to Section 6, Setting Default Contrast and Brightness for LCD Monitor.

2. If **CRT/Custom** displays in the *Display Type* field:

   a. Use the default value of **.95**. If the customer does not want this value, type the desired gamma value into the *Edit* box to the right of the *Display Type* field (shown in Illustration 5-1) and press **<Enter>**.

   b. When a pop-up displays, select **[OK]**. (See Illustration 5-2.)

   c. Click the **[Configure]** button on the Install GUI to save the changes. (If the gamma value was changed, a slight change in the screen intensity will be noticed when Configure is completed.

   d. When a pop-up displays, select **[OK]**.

   e. Select **File** and **Exit** and **[OK]** from the I*nstall/Reconfigure* window.

   f. **Reboot** the system. (A reboot is required to activate the changes.)

   g. This completes the Gamma Calibration process. Proceed to Section 6, Setting Default Contrast and Brightness for LCD Monitor.

6 - SETTING DEFAULT CONTRAST AND BRIGHTNESS FOR LCD MONITOR

This section provides insructions to adjust the monitor to a default condition and proceed with calibrations. Refer to the vendor manual that came with the new LCD for operation of the monitor controls.

1. Open the **Screen Setup**. (Refer to the vendor manual that came with the monitor.)

2. If there is an “Auto Adjust” feature, initiate the auto adjustment of the Phase/Vertical and Horizontal Centering. If it does not, skip to Step 3.

Note

Manual adjustment of the image position to the centerpoint of the LCD screen in all four directions is rarely needed. Auto Adjust takes care of this.

3. Find and set the monitor “Color Temperature” to **(9300)**. Refer to the vendor manual that came with the monitor.
4. Press the **[Exit]** button. (The monitor adjustment is complete, and the LCD monitor is ready for integration to Signa.)

Note

For further information on the setup screens and button identification, refer to the User's Manual that came with the LCD Display.

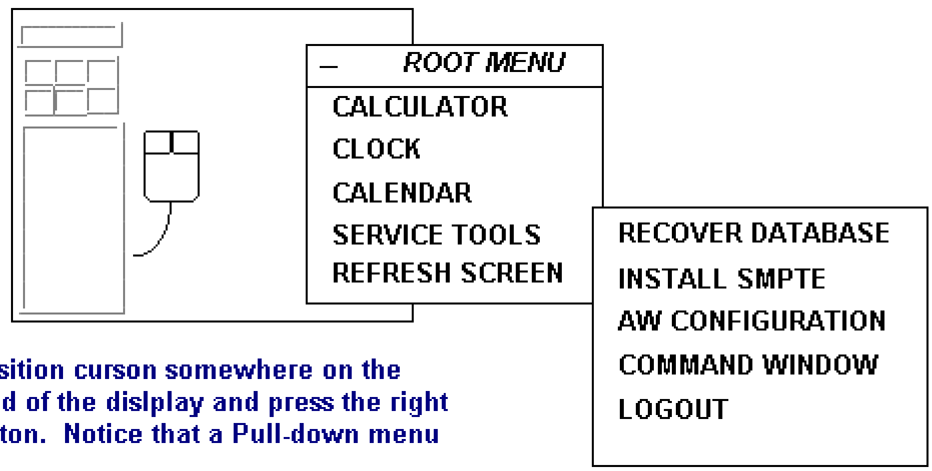
7 - FINAL ADJUSTMENT OF CONTRAST AND BRIGHTNESS

The SMPTE (Society of Motion Picture and Television Engineers) test pattern is used to provide a standard image for calibrating the Window and Level of the display for an MR scanner.

7-1 Displaying SMPTE Pattern

The Window and Level adjustments must be set to ensure site-to-site uniformity in image appearance. The SMPTE test pattern is available on the Operator Workstation Host Computer after IRIX operating system is booted, and you have logged into the system.

1. Install the SMPTE pattern using the steps in Illustration 7-1.

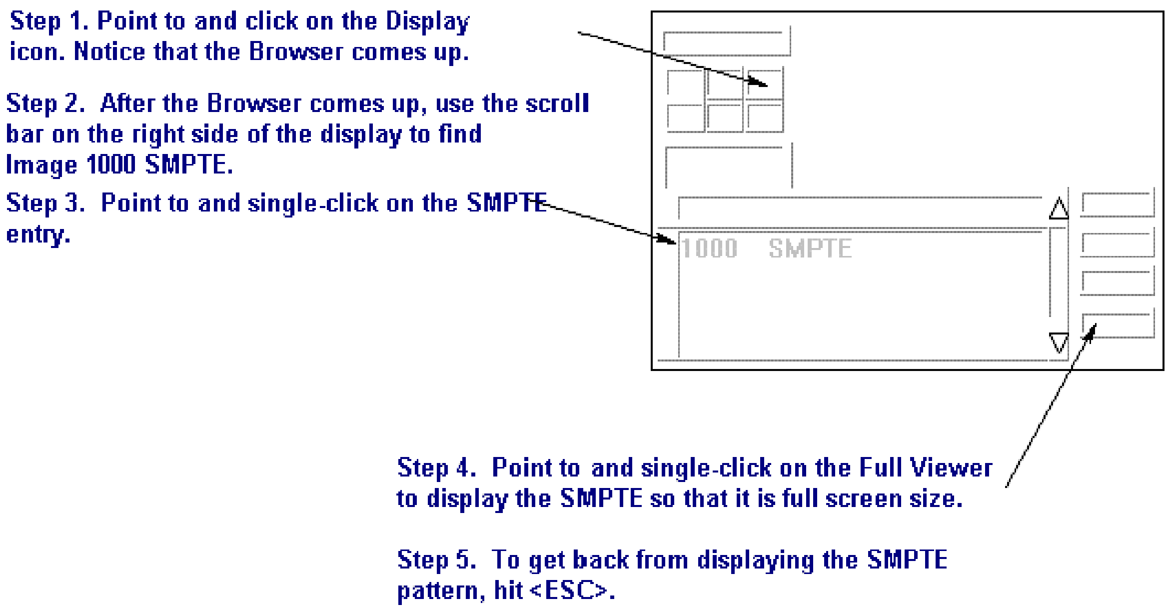


Step 1. Position curson somewhere on the background of the dislplay and press the right mouse button. Notice that a Pull-down menu appears.

Step 2. Slide the mouse down to [Service Tools], and then over to [Install SMPTE]. This will load the SMPTE pattern into the database as Image 1000.

INSTALLING SMPTE PATTERN
ILLUSTRATION 7-1

2. Display the SMPTE pattern by following the steps in Illustration 7-2. Switch to the browser, and select the SMPTE pattern from the patient list to display it. (If necessary, use the *Format* option to select **Full Screen** mode.)



INSTALLING THE SMPTE PATTERN
ILLUSTRATION 7-2

7-2 Adjusting for Proper Window and Level Setting

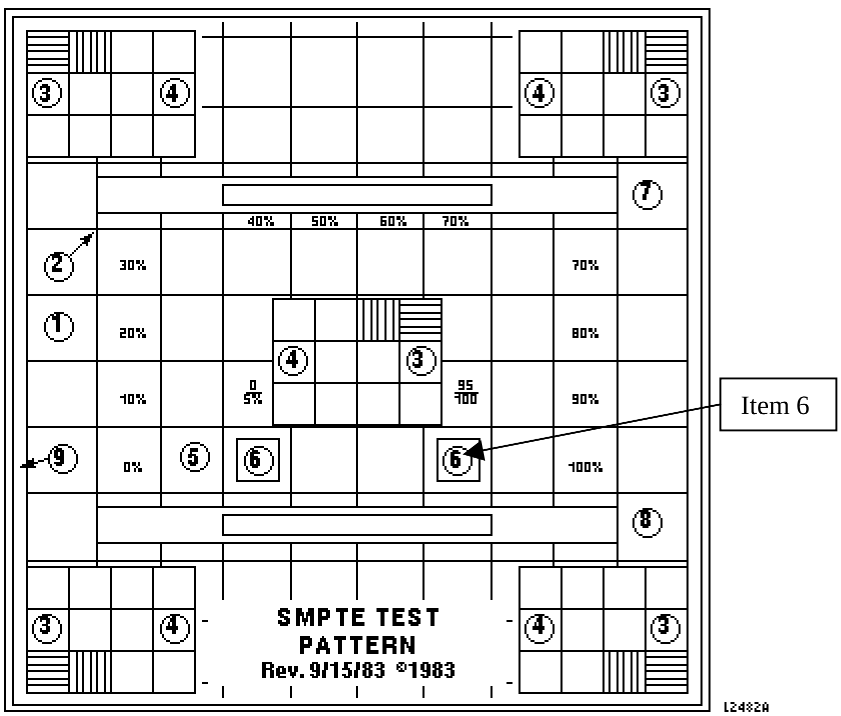
Refer to Illustration 7-3 for the following steps.

1. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse into the horizontal plane.

2. Look at the window control value at the base of the image, and set window control value to **100**.

3. With the cursor inside the displayed image, hold down the middle mouse button and move the mouse into the vertical plane.

4. Look at the level control value at the base of the image, and set level control value to **1024**.



SAMPLE SMPTE TEST PATTERN
ILLUSTRATION 7-3

7-3 Fine Tuning Contrast and Brightness Using SMPTE Pattern

This section provides instructions on the fine adjustments with the Contrast and Brightness controls on the LCD monitor to optimize the SMPTE test pattern for the Window and Level controls. Minimal adjustment should be necessary at this time, but it is important to fine tune the Contrast and Brightness level using an industry standard test pattern. This is an iterative process by adjusting the Contrast, then Brightness, and back to Contrast.

Note

To reduce visible tearing or smearing of the pattern, lower the contrast.

To finely adjust the Brightness and Contrast controls:

1. Look at the SMPTE test pattern and if the 5% and 95% patches are not visible (Item 6 in Illustration 7-3), adjust the brightness control of the display monitor until they are just barely visible.

   (The contrast may also need readjustment if tearing or smearing of the pattern occurs. Refer to Items 1 and 2 and 5 through 8 in Illustration 7-3. Continue adjusting both brightness and contrast until the image is as crisp and clear as possible and the 5% and 95% patches are barely visible.)

2. When satisfied with the display, **exit** the menu and **save** the adjusted Contrast/Brightness settings.

   Note

   These settings will remain in effect even if power is removed or the system is rebooted. Because an operator can tamper with the Brightness and Contrast adjustments, it is recommended that all operators be made aware of the affect they could have on system image quality and the camera.

   If the operator chooses to deviate from this Brightness and Contrast adjustment, ensure the camera imaging and calibration are **not** affected.

3. Remove the SMPTE test pattern by selecting a different Exam from the browser.

8 - CAMERA CALIBRATION

The following procedure describes the steps necessary to verify and set the camera parameters. Once the display is recalibrated, it is essential to recalibrate the camera before the system is used for filming. Although a qualified GE Service Engineer could perform the steps below, it is recommended that this procedure be performed with the on-site assistance of the Camera Vendor Field Engineer.

8-1 DASM Interpolation Setup

1. Select the **<Install>** key from the Service Desktop. When prompted, log in using the password: **operator**.

2. Select the **DASM** folder from the top of the Guided Install GUI once it displays.

3. Set the DASM Interpolation method to **linear**.

4. Before exiting the GUI, insert the **SaveInfo** or new MOD into the drive, go to the top left corner of the Install GUI, and select **File** and **Save GI Configuration to MOD**.

Note

This process does not create a SaveInfo disk. It only creates a copy of the information already entered on the GUI tabs for use in the next software installation.

5. Exit the Install GUI by selecting **File** and **Quit**. (If any error conditions still exist, a warning will display indicating that no changes will be made before exiting.)

8-2 Camera Imaging Look-Up Table

Verify with the Camera Vendor Field Engineer that the currently installed camera look-up table is designed to provide perceivably linear contrast for the light box conditions in the customer’s viewing area. If not, request this FE to replace it with the appropriate look-up table.

8-3 Camera Maximum Optical Density

For optimal reviewing, the light box luminance of the diagnostic region of the film should be in the range of 50 to 500 nits. Vary the maximum Optical Density (OD) setting of the camera to compensate for the light box and to meet this value. A good starting position is a maximum density of 2.8.

Note

The Radiologist performing the image review may refine the final OD settings.

8-4 Camera Contrast

1. With the maximum/minimum optical densities set to compensate for the review area's light box, select a look-up table for the camera that will produce a perceivably linear grayscale for the same light box and the overall ambient light conditions of the viewing area.

Note

The DICOM 3.14 Standard specifies the Barten's curve for linear perception. It is recommended that the manufacturer base the perceptual linearity on this curve.

2. Film the SMPTE test pattern on a 1-on-16 format display. Verify that the 5% and the 95% levels are visually equivalent. If not, perform a Contrast test with the SMPTE test pattern:

   a. Select the new contrast setting from the contrast image set. A good value for the Imation Dry View is **3**.

   b. Use the camera's calibration procedure to set the contrast setting. Ensure that the camera maintains a perceivably linear grayscale.

Note

Filming the SMPTE test pattern for contrast calibration may be optional for the Camera Vendor.

3. Ask the technologist to display a clinical image and set the Window and Level controls for the desired appearance. (A good image to start with is a sagittal or axial head image.)

4. Capture the image on the keypad or host control interface.

5. Print a Contrast test film, and ask the technologist to select the image that best matches the displayed image on the monitor.

6. Record the image number below the selected image, and set the Contrast control to this value.

8-5 Anatomical Filming

This portion of the procedure requires the technologist to verify the camera settings with true anatomical images.

1. Film representative anatomical images to confirm the settings. The image set should include T1 and T2 head images, joint images and c-spines.

2. Observe the accuracy of the low tones, mid tones and high tones. If a filmed image is found nonequivalent, recalibrate the camera based on the customer's evaluation.

9 - TROUBLESHOOTING GUIDE

The following section provides suggestions for troubleshooting the LCD monitor setup.

9-1 No Picture

Make sure:

- Signal cable is completely connected to the display card/computer.
- Display card is completely seated in its slot.
- Power Switch and computer power switch are in the ON position.
- Signal cable connector does not have bent or pushed in pins.
- A wrong data cable does not exist on the AWS; if a SUN Adapter is installed on the data cable, remove it.

9-2 Image Persistence

Image persistence occurs when a ghost of an image remains on the screen even after the monitor is turned off. Unlike CRT monitors, the LCD monitor's image persistance is not permanent. To alleviate image persistance, turn off the monitor for as long as an image was displayed. If an image was on the monitor for one hour and a ghost of that image remains, the monitor should be turned off for one hour to erase the image.

Note

It is recommended that a screen saver be used whenever the screen is idle.

9-3 Camera Film Changed After Monitor Replacement and/or Gamma Calibration

Make sure that the Monitor and Camera Calibration procedures were performed, and the correct LUT or Gamma value installed for the monitor.

If there is no remote camera in the list, the *Film Composer* menu cannot be accessed. Try to set the camera to **Replicate** instead of the linear mode.

For Kodak cameras, set the camera to **Bi-linear** mode:

1. Open the Image browser.
2. Open **Film Composer**.
3. Select **Configure**.
4. Select the remote printer being used by the site.
5. Select **Update**.
6. Under magnification, select the **replicate** option (for Kodak: Bi-linear) and **OK**.
7. Select **Done**.
8. Select **Quit**.
9. Select **OK**.
10. Return to original Desktop, and check the film and adjust camera if necessary.

9-4 Image Unstable, Unfocused or Swimming

Make sure:

- Signal cable is completely attached to the computer.
- Signal timings of the monitor and display card.
- For LCDs only – To change the video mode to non-interlace and if possible use a 50 Hz refresh rate.
- To adjust the monitor controls to focus and adjust display. (Refer to the vendor manual that came with the LCD Monitor for the operation of this function.)
- For LCDs only – To change the video mode to non-interlace if text is garbled, and use a 50 Hz refresh rate if possible.

9-5 LED on Monitor Unlit

Make sure:

- Power Switch is in the ON position and Power Cord is connected.
- Computer is not in a power saving mode.

9-6 Display Image Sized Improperly

Push the [Auto Adjustment] button. (Refer to the vendor manual that came with the LCD Monitor for the operation of this function.)

9-7 Selected Resolution Displayed Improperly

Push the **[Auto Adjustment]** button. (Refer to the vendor manual that came with the LCD Monitor for the operation of this function.)

9-8 Diagnostic Image Quality Traced to Problem with LCD Setup

Operator has adjusted the Brightness and Contrast of the monitor, which has affected Camera/Film imaging. Recalibration may be necessary. Start with Section 1 of this procedure.

9-9 Display Image Has Green or Red Cast

Make sure the *Color Temperature* is set to **9300**. (Refer to the vendor manual that came with the LCD Monitor for the operation of this function.)

9-10 Display Image Ghosts or Gray Lines in Areas of Extreme Black & White Transitions

The LCD Refresh Rate is too fast and must be slowed down to allow for the pixels to react properly. To resolve this issue, all LCD monitors should be set to Refresh Rate (also known as a Line Rate or Frame and Swap Rate) of 60 Hz. See Section 4 of this procedure.

9-11 Display Image Blurry or Blotchy (Images Described as Poor Grayscale)

LCDs are highly dependent on gamma correction. Perform the gamma correction procedure in Section 5 to solve this problem.

The off-the-shelf LCDs on Signa are optimized for PC use from the factory, but **not** Diagnostic Imaging. The LCD monitors are **not** plug and play with regard to the Signa applications software and SGI computer video system. This requires that a complete monitor setup and calibration be performed at:

- Installation,
- During a replacement of a failed monitor, or
- Any time software is reloaded.

9-12 Customer Complaints of Poor Grayscale, Window or Level, Lack of Definition

LCD technology uses three-color film plates and a backlight to display different shades of color and gray. They are highly dependent on gamma correction. The off-the-shelf LCDs on Signa are optimized for PC use from the factory, but **not** Diagnostic Imaging. Attaching an LCD to the Signa system without calibrating it will cause image quality problems that can be interpreted as system problems, leading to unnecessary:

- Hardware replacement,
- Protocol Troubleshooting, or
- Customer Satisfaction Issues.

LCD monitors are **not** plug and play with regard to the Signa applications software and SGI computer video system. This requires that a **complete** monitor setup and calibration be performed at:

- Installation,
- During a replacement of a failed monitor, or
- Any time Software is reloaded.

9-13 LCD Facts

Be aware of the following:

- Monitor calibration information is **not** saved during the SavInfo backup routine or accessible via the Health page.

- There is a big difference between CRT and LCD monitors. CRTs are much less sensitive to a wrong gamma setting.
- Gamma Tables (Look-Up Tables or LUTs) are highly specific to the LCD model, and should only be applied to the correct model they were designed for.

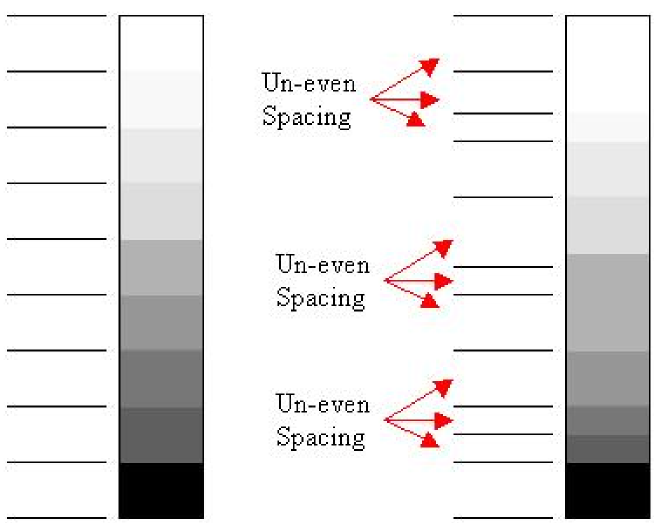
- Gamma Tables consist of thousands of luminescent data points that are used by the electronic circuitry to generate uniform transitions from one color or grayscale level to the next.

- All recent LCDs used in Signa have a best fit or single gamma value, as well as a complete Gamma Table to maximize flexibilty when setting up the monitor.

- It is recommendd that the Gamma Table always be used as a default.

9-14 What is Gamma?

Simply stated, gamma normalizes the grayscale so each level is uniform in size and shape across the entire display. (See Illustration 9-1 for a grayscale comparison.)



GRAYSCALE COMPARISON
ILLUSTRATION 9-1

Without getting into complicated equations for gamma correction, adjust the grayscale or color gradient so each “step” is exactly the same size and shade of color or gray. If the steps are not uniform in size and color, the monitor will display the error as poor transitions from one step to the next. This appears to the eye as small ghosts or ragged edges.

9-15 How to Set Correct Gamma

When the Monitor Type is selected in the Install GUI, the correct gamma table (LUT) for that monitor is **automatically loaded**. The gamma LUT remains active even if a reboot occurs.

9-16 Can Gamma Command Be Used to Set Gamma?

No. This command is no longer an effective way to change gamma permanently. The gamma command should only be used if there is a need to temporarily change to a single gamma value, overriding any previously installed Gamma LUT.

Using the gamma command on systems with an LCD will most likely cause image quality problems. Using a Gamma Table will produce a better image quality in most cases rather than a single gamma value. A single gamma value is much less effective at providing a uniform grayscale on LCDs.

9-17 How to Determine if Correct Gamma Table or Setting Is Installed

Open up the Install GUI and look at the *Display Type* field to the right of the *Option* box. It will either display a monitor type or a single number.

- If a Monitor Type is in this box, a Gamma Look-Up Table (LUT) is installed for that monitor.
- If a Single Value displays, the CRT/Custom Option was used to set a Single Gamma value.

9-19 When “gamma” Typed in C-Shell, Is Displayed Number the Gamma Setting?

No. It is a meaningless number left over from earlier software versions. A gamma table is installed by selecting the Monitor Type in the Install GUI.

9-20 When “gamma 1.05” Typed in C-Shell, Why Does LCD Change in Appearance?

IRIX operating system commands, such as “gamma," take priority over the Signa applications software. The Install GUI gamma tools are written in IRIX script and have less priority. Since the O/S gamma command was envoked with a value, that value will now override the Applications software. The single value of 1.05 temporarily overrides the gamma LUT until a **reboot** occurs. Upon reboot, the gamma LUT resumes control.

9-20 Can a Gamma Value of 1.05 Used Previously Still be Used?

Do **not** use the gamma value of 1.05. All LCD and CRT monitors now use **.95**, which was determined to be the best default single value. Using the gamma command on systems with an LCD will most likely cause image quality and filming problems. Using a Gamma Table rather than a single gamma value (such as 1.05) will in most cases produce less image quality problems than a single gamma value. A single gamma value is much less effective at providing a uniform grayscale on LCDs.

9-21 Customer Dislikes Gamma Setting Provided by Gamma Table

Change it using the Install GUI. Under Display Type, select **CRT/Custom** and type in a single gamma value that you and the customer agree upon. A reboot will make the single setting permanent.

There is a single optimized "nominal" gamma value for each monitor type, as well as a gamma table. Consult the Troubleshooting section within the proper monitor calibration procedures to find the nominal value for the monitor type. Using a single value affects the camera to a higher degree than a Gamma Table, so be prepared to adjust the camera as well.

9-22 Single Value Optimization Technique

To minimize the time needed to find a single gamma value acceptable to the customer, the IRIX O/S gamma command can still be used in a C-shell to tweak a single gamma value as follows:

- Type **gamma n.nn** (where *n.nn* is the nominal single gamma value for the monitor type) will generate an immediate change to the monitor. (Increasing the value will increase brightness and contrast together. Decreasing it will have the opposite effect).

- Continue to adjust the gamma setting, checking images and films until an acceptable setting is found, and record the final value.

- **Remember:** To make the value permanent, open the Install GUI, select **CRT/Custom** under Display Type, and type in the new single gamma value.

- Reboot the system to make this the permanent setting.

Note

If a C-shell is opened and "gamma" typed, the number may not match what was installed in the Installation GUI. This is normal.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASON FOR CHANGE
0	Feb 26, 2003	D. Hofstetter	Initial version. This is a “Generic” LCD Monitor setup procedure.
1	Mar 5, 2003	D. Hofstetter	Added Troubleshooting hints.
2	May 12, 2003	D. Hofstetter	Added Line Rate script description.
3	June 4,2003	D. Hofstetter	Modified Line Rate adjustment based on O/S version.
4	July 9, 2003	D. Hofstetter	Added Process for loading gamma using the patch cdrom
5	July 25, 2003	D. Hofstetter	Removed reference to old gamma LUT installation.
6	Oct 6, 2008	P. Burback	Table 1-1: Added link to NEC 1990SX Monitor Calibration and Gamma Setup (OW2SCA6H.doc) per PQR 13153333.